

Shahjalal University of Science & Technology
Department of Computer Science & Engineering
2nd Year 1st Semester Lab Final Examination (Session 2021-2022)
Course: **CSE 234** Course Title: **Object Oriented Programming Language Lab**
Credits : **2.0** Full Marks : **60** Time : **1 Hour 20 Min**

Instructions:

1. If you have a **phone**, **USB flash drive**, or any digital **device** with you, keep them in your bag. If you are found with any of these items later you will be **expelled**.
2. Create a **folder** on the **desktop** named after your **registration number** and put all the code files in it. If your answer code files are **not found** on the **desktop** then your answer codes will **not be taken**.
3. At the start of each source **code file** write your **name**, and **registration** number in the **comment**.
4. If you need to create package folders those also should be created inside this folder.
5. All class, method, and variable **names** that are provided should be **kept the same** (case-sensitive) as it is.
6. Keep your code **font size small** enough in **VSCode** at all times so that they are not readable to nearby students.
7. Any type of **communication** (verbal or digital) with peers is strictly **prohibited**. If you try to see/show each other's code or try to talk, you will be **expelled**.
8. Whenever exam time is finished, you should close all opened files/applications and **turn off** your **monitor**. **Do not shut down your computer**.

Q#1

Marks: 20

The following file **BQ1.java** is provided to you. You cannot change any code in this file.

```
1 // BQ1.java
2 abstract class A { int i; A(int i) {this.i = i;}
3     int m(int j) { return 2*i+j; } abstract int m2(int j, int k); void m4(){}
4 class B extends A { B(int i) { super(3*i);}
5     int m2(int x, int y) { return x*y; }
6 }
7 public class BQ1 {
8     public static void main(String[] args) {
9         C c = new C(10, 20); B b = new B(5); System.out.println("Main B Start");
10        c.m3("Java"); System.out.println("1:"+c.m2(6,6));
11        System.out.println("2:"+b.m2(6,6));
12        A a = b; System.out.println("3:"+a.m(15)); System.out.println("4:"+a.i);
13        a = c; System.out.println("5:"+a.m(15)); System.out.println("6:"+a.i);
14        a.m4(); c.i+=115; a.m4(); a.i+=210; a.m4(); new C(100,200);
15    }
16 }
```

Now write a Java Class named “C” in the **C.java** file, so that if we run the **main** method in **BQ1** it gives the following output:

Exam B Started
Data B Loaded
Main B Start
Java<>Java
1:24
2:36
3:45
4:15
5:35
6:10
20
135
345
Data B Loaded

Consider the following class **BQ2** and the **main** method inside it. The following codes are provided to you in the **BQ2.java** file. You cannot change any code in this file.

```
1 // BQ2.java
2 package pkB;
3
4 class S implements pkB.p3.PC.IPC2 {}
5
6 public class BQ2 {
7     public static void main(String[] args) { System.out.println("Start");
8         pkB.p1.PA pa = new pkB.p1.PA(); System.out.println(pa.m("MNR"));
9         System.out.println(pa.m4(8)); System.out.println(pa.m4(3));
10        pkB.p3.PC.IPC ipc = pa; System.out.println(ipc.m4(15));
11        System.out.println(pkB.p1.p2.PB.m2("ABCD"));
12        pkB.p1.p2.PB pb = new pkB.p1.p2.PB(); System.out.println(pb.m(4,2));
13        System.out.println(pb.m(10,20));
14        pkB.p3.PC pc = pb; System.out.println(pc.m(4,6));
15        System.out.println(pkB.p3.PC.m3("TBD"));
16        S s = new S(); System.out.println(s.m2(9));
17        pkB.p3.PC.IPC2 ipc2 = s; System.out.println(ipc2.m2(20));
18    }
19 }
```

Now write a Java Class named “PA” in the **PA.java** file, “PB” in the **PB.java** file, and, Interface named “PC” in the **PC.java** file according to their required packages, so that if we run the **main** method in **BQ3** class it gives the following output:

Output:
Start
(MNR)
48
18
90
ABCDABCDABCD
24
600
72
TBD***TBD
90
200

Consider the following class **BQ3** and the **main** method inside it. The following codes are provided to you in the **BQ3.java** file. You cannot change any code in this file.

```

1 // BQ3.java
2 public class BQ3 {
3     public static void main(String[] args) {
4         for (int n=5;n>-1;n--) {
5             System.out.println("Exceptions Case: " + n);
6             try {
7                 if (n == 0) { System.out.println("Starting: "+n);
8                     Exc.a(); System.out.println("Ending: "+n); }
9                 if (n == 1) { System.out.println("Starting: "+n);
10                    Exc.b(); System.out.println("Ending: "+n); }
11                try { if (n == 2) { System.out.println("Starting: "+n);
12                    Exc.c(); System.out.println("Ending: "+n); }
13                    else if (n == 3) { System.out.println("Starting: "+n);
14                        Exc.d(); System.out.println("Ending: "+n); }
15                    else { System.out.println("Starting: "+n);
16                        throw new NewExc(n); }
17                } catch (ArrayIndexOutOfBoundsException ex) {
18                    System.out.println("Exception is "+ex);
19                    if(ex.getCause()!=null) System.out.println("Cause: "+ ex.getCause());
20                }
21            } catch (ArithmeticException | NullPointerException ex) {
22                System.out.println("Exception = "+ex);
23                if(ex.getCause()!=null) System.out.println("Cause: "+ ex.getCause());
24            } catch (NewExc ex) {
25                System.out.println(ex);
26                if(ex.getCause()!=null) System.out.println("Cause: "+ ex.getCause());
27            } catch (Exception ex) {
28                System.out.println("Any Exception");
29                if(ex.getCause()!=null) System.out.println("Cause: "+ ex.getCause());
30            }
31        }
32        System.out.println("Exam Ended");
33    }
34 }

```

Now write a Java Class named “**Exc**” in the **Exc.java** file, and, “**NewExc**” in the **NewExc.java** file so that if we run the **main** method in **BQ3** class it gives the following output:

Output:

```

Exceptions Case: 5
Starting: 5
New B Exception: New B Exception: 5
Exceptions Case: 4
Starting: 4
New B Exception: New B Exception: 4
Exceptions Case: 3
Starting: 3
New B Exception: New B Exception: 0
Exceptions Case: 2
Starting: 2
Exception is java.lang.ArrayIndexOutOfBoundsException
Cause: java.lang.NullPointerException
Exceptions Case: 1
Starting: 1
Exception = java.lang.NullPointerException: No Data!
Exceptions Case: 0
Starting: 0
Exception = java.lang.ArithmeticException
Exam Ended

```